AP BIOLOGY · UNIT 1 · TOPIC 1.3

Introduction to Macromolecules

Building Up & Breaking Down: The Chemical Reactions Behind Biological Macromolecules

What You'll Learn

This set of study slides is aligned to **AP Biology Unit 1, Topic 1.3** and focuses on the two fundamental chemical reactions that govern how macromolecules are assembled and disassembled in living systems.

1

Learning Objective

Describe the chemical reactions that **build** and **break** biological macromolecules, including the role of water in each process.

2

Essential Knowledge

Understand **dehydration synthesis** (condensation) and **hydrolysis** — the two complementary reactions that connect and disconnect monomers to form or break polymers.

3

Key Vocabulary

Monomer, polymer, covalent bond, hydrolysis, dehydration synthesis, polymerization, hydrogen ion (H^+), hydroxyl group (OH^-).

Big Picture: Monomers, Polymers & Why It Matters

All four classes of biological macromolecules — **carbohydrates, lipids, proteins, and nucleic acids** — are built from smaller subunits called **monomers**. When monomers link together through covalent bonds, they form long chains called **polymers**. This process of connecting many monomers is known as **polymerization**.

Why does this matter for AP Bio? Because every time your cells need to grow, repair damage, store energy, or read genetic information, they must either **build new polymers** or **break existing ones down**. Two elegantly simple — yet critically important — reactions make this possible:

- **Dehydration synthesis** (also called condensation) — joins monomers together by removing water
- **Hydrolysis** — breaks polymers apart by adding water

Notice the pattern: **water is the common thread**. One reaction removes it; the other adds it. Understanding these two reactions is the foundation for everything else in Unit 1.

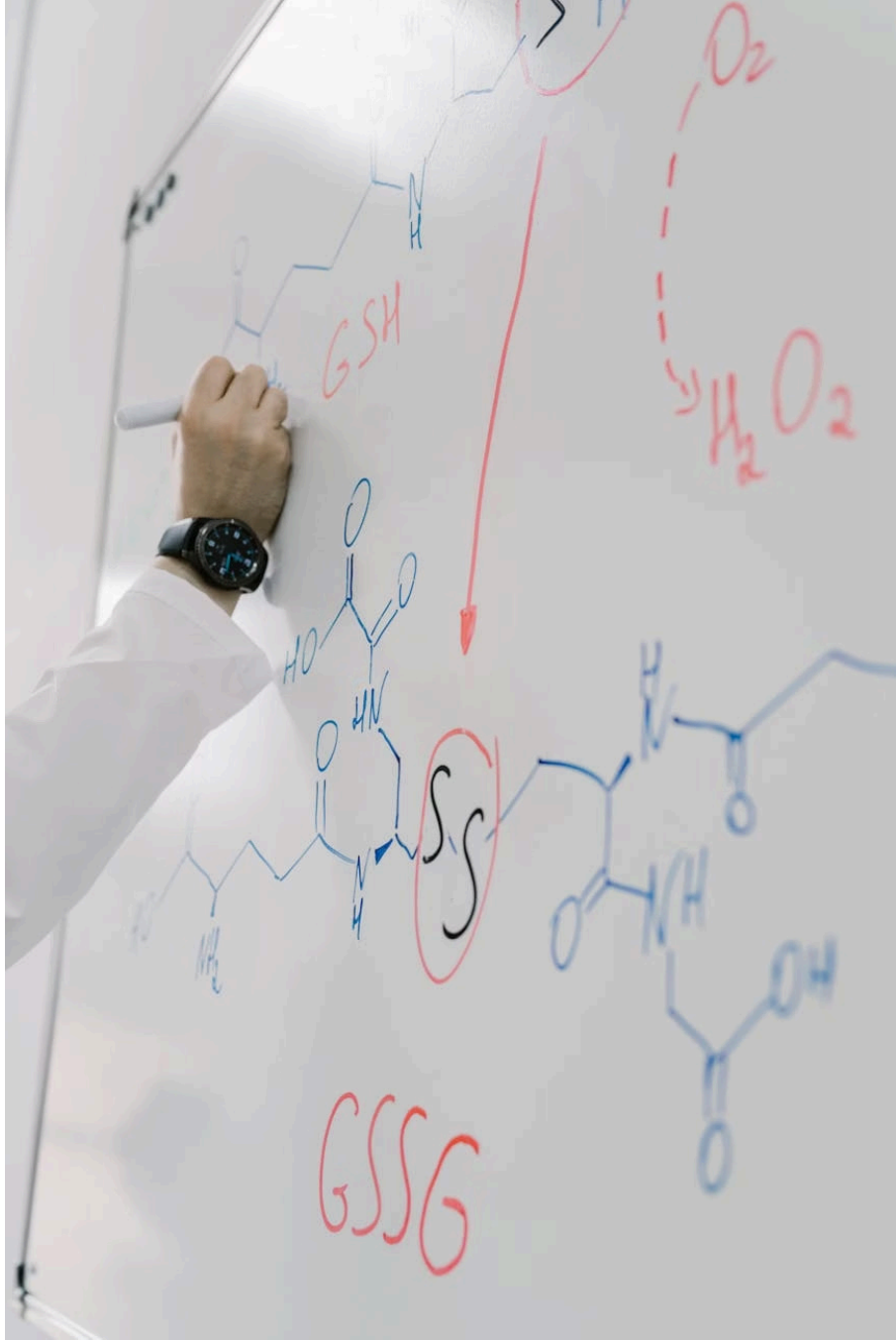
🔑 **AP Exam Tip:** The College Board frequently tests whether you can identify the *direction* of each reaction. Remember — dehydration synthesis **releases** water and builds polymers, while hydrolysis **consumes** water and breaks polymers. Confusing the two is one of the most common mistakes on the exam.

CHAPTER 1

REACTION #1

Dehydration Synthesis: Building Polymers

How cells connect monomers into the macromolecules that power life



How Dehydration Synthesis Works

Dehydration synthesis (also called a **condensation reaction**) is the process by which two smaller molecules — typically monomers — are **joined together through the formation of a new covalent bond**. The "dehydration" part of the name tells you exactly what happens: water is *lost* from the reactants during the reaction.

Here is the step-by-step mechanism in detail:

1 Hydrogen Removed

A **hydrogen ion (H)** is removed from a functional group on **Monomer A**. This leaves Monomer A with a reactive site ready to bond.

2 Hydroxyl Removed

A **hydroxyl group (–OH)** is removed from **Monomer B**. Together, the removed H and –OH combine to form a molecule of water (H_2O).

3 Covalent Bond Forms

The two monomers are now covalently bonded, forming a **dimer**. Repeating this process with additional monomers builds a polymer — a process called **polymerization**.

Polymerization: Repeating Dehydration Synthesis

When dehydration synthesis is repeated over and over, monomer after monomer is added to a growing chain. This repetitive process is called **polymerization**, and it is how cells manufacture every polymer in your body — from the **starch** in a potato to the **DNA** in your chromosomes.

Each time a new monomer is added:

- One **water molecule** is released as a byproduct
- One new **covalent bond** is formed
- The polymer grows by exactly **one monomer unit**

This means a polymer made of 100 monomers required **99 dehydration synthesis reactions** and produced **99 water molecules** during its assembly. That's a lot of water — and it highlights why cells must be in an aqueous (water-rich) environment to function properly.

In living cells, dehydration synthesis reactions are **endergonic** (they require energy input) and are facilitated by **enzymes** — specialized protein catalysts that lower the activation energy of the reaction and ensure it proceeds quickly and accurately.

Real-World Examples

- **Amino acids → Polypeptide** (peptide bonds)
- **Monosaccharides → Polysaccharide** (glycosidic linkages)
- **Nucleotides → Nucleic acid** (phosphodiester bonds)
- **Fatty acids + Glycerol → Lipid** (ester bonds)

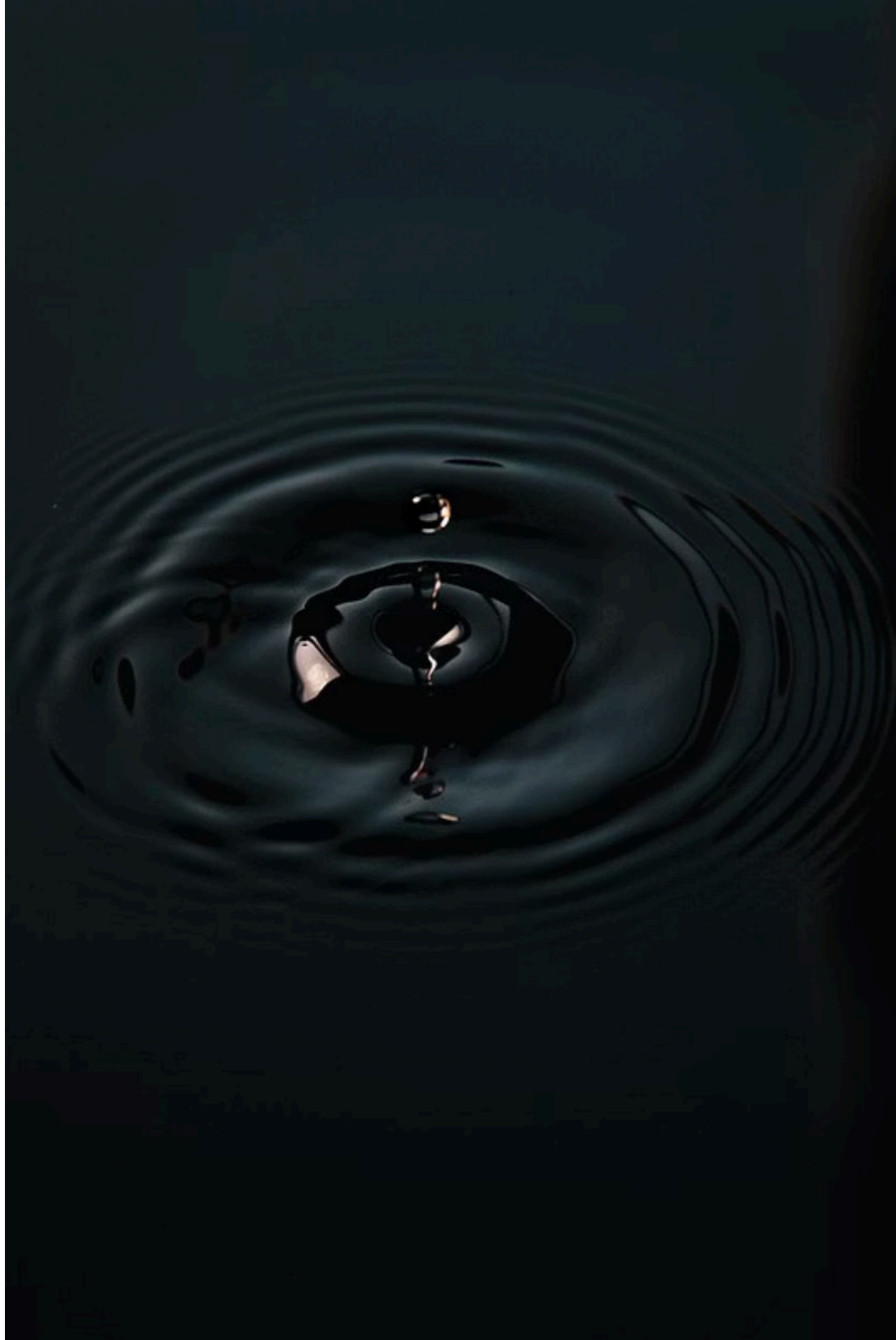
Each of these examples follows the same fundamental pattern: remove H from one subunit, remove –OH from the other, release H_2O, form a covalent bond.

CHAPTER 2

REACTION #2

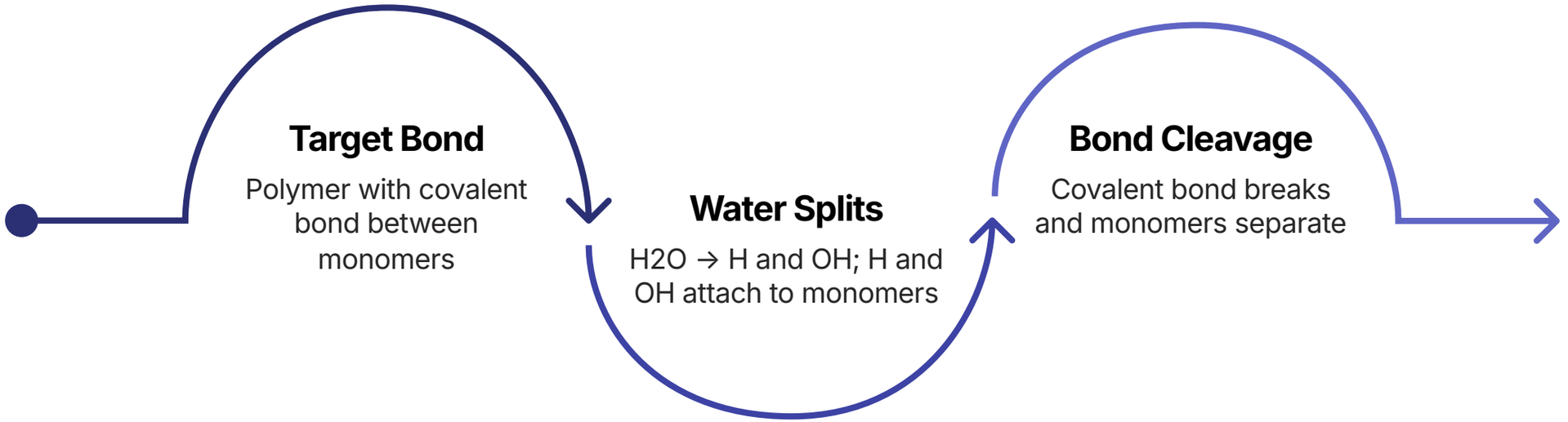
Hydrolysis: Breaking Polymers

How cells disassemble macromolecules by adding water to covalent bonds



How Hydrolysis Works

Hydrolysis is essentially the **reverse** of dehydration synthesis. The word itself comes from the Greek *hydro* (water) + *lysis* (to break apart). In a hydrolysis reaction, **water is added to a covalent bond between monomers in a polymer**, and that bond is broken — splitting the polymer into smaller molecules.



Here is the detailed mechanism:

1 Water Molecule Splits

A water molecule (H_2O) is split into its two components: a **hydrogen ion (H)** and a **hydroxyl group (–OH)**.

2 Components Are Added

The **hydrogen ion (H)** is added to one monomer, and the **hydroxyl group (–OH)** is added to the adjacent monomer. This satisfies the bonding requirements of each monomer.

3 Covalent Bond Breaks

The covalent bond between the two monomers is **cleaved** (broken), releasing the individual monomers as separate, complete molecules ready to be recycled or used for energy.

Hydrolysis in Your Body: Real Examples

Hydrolysis is happening inside your body **right now**. Every time you digest food, your digestive enzymes catalyze hydrolysis reactions that break down the macromolecules in your meal into absorbable monomers. Here are some key examples:

Digesting Carbohydrates

Enzymes like **amylase** (in your saliva and small intestine) hydrolyze the glycosidic linkages in **starch**, breaking the polysaccharide into individual **glucose** monomers that can be absorbed into your bloodstream and used for cellular respiration.

Digesting Proteins

Proteases such as **pepsin** (in stomach acid) and **trypsin** (in the small intestine) hydrolyze peptide bonds, breaking polypeptide chains into individual **amino acids**. These amino acids are then reassembled into your body's own proteins via dehydration synthesis.

Breaking Down Nucleic Acids

Nucleases hydrolyze the phosphodiester bonds in DNA and RNA, releasing individual **nucleotides**. This is critical during processes like DNA repair, apoptosis (programmed cell death), and the recycling of cellular components.

Side-by-Side: Dehydration Synthesis vs. Hydrolysis

These two reactions are mirror images of each other. Understanding how they compare is one of the most important concepts in AP Bio Unit 1. Study the comparison below carefully:

Feature	Dehydration Synthesis	Hydrolysis
Also called	Condensation reaction	Hydrolytic cleavage
What happens to water?	Water is **produced** (released)	Water is **consumed** (used up)
What happens to monomers?	Monomers are **joined** together	Polymer is **broken** into monomers
Bond change	Covalent bond is **formed**	Covalent bond is **broken**
H and –OH movement	H removed from one monomer; –OH removed from the other	H added to one monomer; –OH added to the other
Energy requirement	Endergonic (requires energy)	Exergonic (releases energy)
Biological role	Building macromolecules (anabolism)	Breaking down macromolecules (catabolism)
Example enzymes	DNA polymerase, ribosome	Amylase, protease, lipase, nuclease

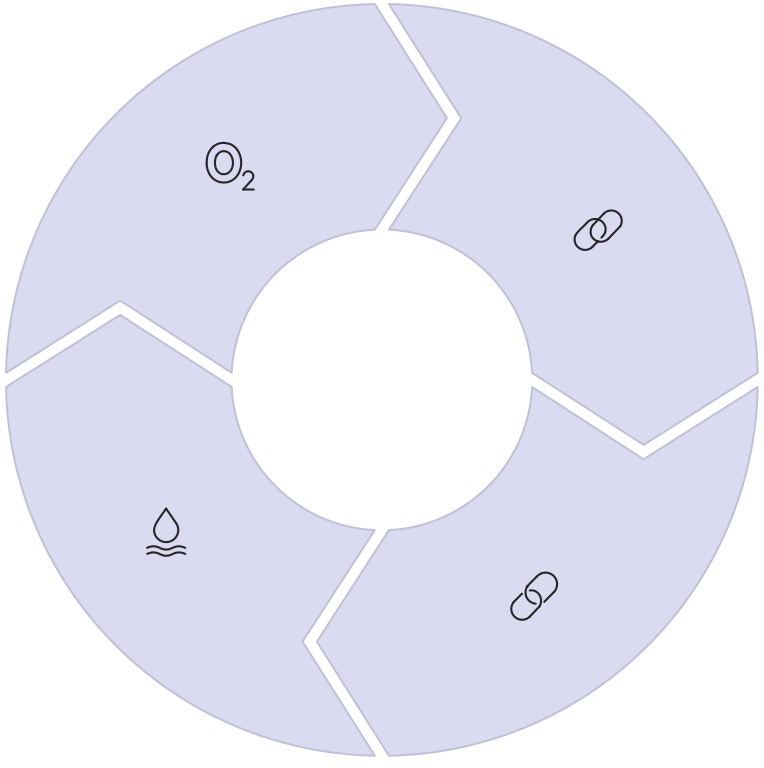
💡 **Memory Trick:** Think **"Dehydration = De-water = water leaves"** and **"Hydro-lysis = water-breaking = water breaks the bond."** If you can remember which direction the water moves, you can reconstruct the entire mechanism on the exam.

Visualizing the Complete Cycle

Dehydration synthesis and hydrolysis form a continuous cycle in living organisms. Cells constantly build and break macromolecules depending on their needs — growth, repair, energy production, or signaling. This dynamic balance is a hallmark of life.

Free Monomers

Individual monomers (amino acids, glucose, nucleotides) are available in the cell's cytoplasm, obtained from food or recycled from old polymers.



Dehydration Synthesis

Enzymes catalyze the removal of H and –OH from adjacent monomers. A water molecule is released and a new covalent bond is formed.

Hydrolysis

When the polymer is no longer needed (or must be digested), enzymes add water to the covalent bonds, cleaving them and releasing individual monomers.

Polymer Formed

Repeated bonding produces a functional polymer — a protein, polysaccharide, or nucleic acid — ready to carry out biological work.

This cycle is happening trillions of times per second across all the cells in your body. It is the molecular basis of **metabolism** — the sum of all chemical reactions in a living organism.

Key Takeaways & Exam Review

Water Is Central

Both reactions revolve around water. Dehydration synthesis **removes** water to form bonds; hydrolysis **adds** water to break bonds. Always track where the H_2O goes.

H + OH = H_2O

In dehydration synthesis, H comes from one monomer and –OH from the other to form water. In hydrolysis, water splits into H and –OH, which are added back to monomers.

Polymerization

Repeating dehydration synthesis builds long polymer chains. A polymer of n monomers requires $n - 1$ reactions and produces $n - 1$ water molecules.

Covalent Bond Cleavage

Hydrolysis specifically targets **covalent bonds** between monomers. The addition of water provides the atoms needed to "cap" the broken ends of each monomer.

AP Exam Reminder: You may be asked to identify whether a diagram shows dehydration synthesis or hydrolysis, predict the products of each reaction, or explain how enzymes facilitate these processes. Practice with diagrams and be ready to describe the movement of H and –OH in your free-response answers.